Summary for Midterm 1

The first midterm tests the material covered till 2/21.
We discussed two general estimation methods:

- the method of moments
- the maximum likelihood method

You need to be able to describe and to implement both methods.

An estimator $\widehat{\theta}$ of the parameter θ is a function of the data and therefore random. If $\mathbb{E}_\theta[\widehat{\theta}^2] < \infty$, we defined its **mean squared error** as

$$\mathrm{MSE}(\widehat{\theta}) = \mathbb{E}_\theta[\widehat{\theta} - \theta]^2 = (\mathbb{E}_\theta[\widehat{\theta}] - \theta)^2 + \mathrm{Var}_\theta(\widehat{\theta}).$$

We call $\widehat{\theta}$ **unbiased** if $\mathbb{E}_\theta(\widehat{\theta}) = \theta$ for all θ;
We call $\widehat{\theta}$ **asymptotically unbiased** if $\lim_{n\to\infty} \mathbb{E}_\theta(\widehat{\theta}) = \theta$ for all θ.

We also discussed **consistency**. An estimator $\widehat{\theta}$ is a **consistent** estimator of θ, or, equivalently we say $\widehat{\theta}$ **converges in probability** to θ, iff

$$\lim_{n\to\infty} \mathbb{P}_\theta\{|\widehat{\theta} - \theta| < \varepsilon\} = 1$$

for all $\varepsilon > 0$. This is the case when $\lim_{n\to\infty} \mathrm{MSE}(\widehat{\theta}) = 0$. [Note that this is a sufficient condition only.]

The **continuous mapping theorem** states that
If $T_n = T(X_1, \ldots, X_n)$ converges in probability to a constant μ, and g is a continuous function, then $g(T_n)$ converges in probability to $g(\mu)$.

This result can be used to prove consistency of an estimator $\widehat{\theta} = g(T_n)$ of the quantity $\theta = g(\mu)$. For instance, let $X_1, \ldots, X_n$ be independent, identically distributed with $\mathbb{E}|X| < \infty$. The law of large numbers states that $\overline{X}_n := \sum_{i=1}^n X_i/n$ converges in probability to $\mu := \mathbb{E}[X]$. Hence, by the continuous mapping theorem, $1/\overline{X}_n$ converges in probability to $1/\mu$, provided $\mu \neq 0$.

The **delta-method** states that
If $T_n = T(X_1, \ldots, X_n)$ has asymptotically a normal $N(\mu, \sigma^2/n)$ distribution, that is, the distribution of $\sqrt{n}(T_n - \mu)$ converges to a $N(0, \sigma^2)$ as $n \to \infty$, and g is a continuously differentiable function at μ, then $g(T_n)$ has asymptotically a normal $N(g(\mu), [g'(\mu)]^2\sigma^2/n)$ distribution for large values of the sample size n.

For instance, let $X_1, \ldots, X_n$ be independent, identically distributed with mean $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2$. Then, the Central Limit Theorem informs us that the distribution of $\overline{X}_n$ is asymptotically $N(\mu, \sigma^2/n)$ and, moreover, we have for any g, that is continuously differentiable at μ, the **distribution** of

$$g(\overline{X}_n) \text{ is approximately normal } N(g(\mu), [g'(\mu)]^2\sigma^2/n)$$

for large values of the sample size n.

The ***distribution* of the maximum likelihood estimator** $\widehat{\theta}$ is in many cases[1] **approximately normal** $N\left(\theta, 1/\{nI(\theta)\}\right)$ for large values of the sample size n.
Here the *Fisher information* $I(\theta)$ is given by

$$\mathbb{E}_\theta \left(\frac{\partial}{\partial \theta} \log f(X|\theta) \right)^2 .$$

A useful formula is

$$\mathbb{E}_\theta \left(\frac{\partial}{\partial \theta} \log f(X|\theta) \right)^2 = -\mathbb{E}_\theta \left(\frac{\partial^2}{\partial \theta^2} \log f(X|\theta) \right),$$

especially since the term on the right is often easier to compute.

We discussed confidence intervals for a parameter θ. We discussed two types:

- *Approximate* confidence intervals based on the large sample (normal) distribution of the maximum likelihood estimator $\widehat{\theta}$. This $100(1-\alpha)\%$ confidence interval has the form
$$\widehat{\theta} \pm z(\alpha/2) \frac{1}{\sqrt{nI(\widehat{\theta})}}.$$
It is of the form $\widehat{\theta} \pm z(\alpha/2)\widehat{s}$, where $\widehat{s}$ is the estimated standard error of $\widehat{\theta}$.
- *Exact* confidence intervals.

The χ^2_n **distribution** is defined as follows: Let $Z_1, \ldots, Z_n$ be iid $\sim N(0,1)$. Then the random variable

$$Y := \sum_{i=1}^n Z_i^2$$

has a χ^2_n distribution. We say Y has a chi-square distribution with n degrees of freedom. We proved that a χ^2_n distribution is the same as a Gamma$(n/2, 1/2)$ distribution. This means in particular that Y has mean

$$\mathbb{E}[Y] = \frac{n/2}{1/2} = n$$

and variance

$$\mathrm{Var}(Y) = \frac{n/2}{(1/2)^2} = 2n.$$

Finally, we argued that for $X \sim \chi^2_n$ and $Y \sim \chi^2_m$, and X and Y are independent, the sum $X + Y \sim \chi^2_{n+m}$.

[1] But not always, one needs *regularity conditions* on $f(x|\theta)$.